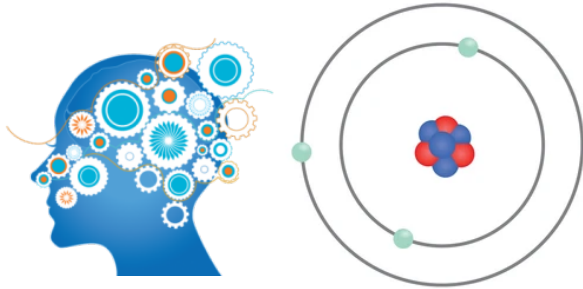


Chromosomes, genes and DNA

Textbook pages: 243 - 246

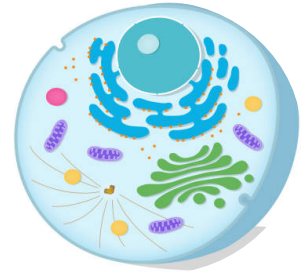
Lesson objectives

- ❑ To explain what chromosomes are made up of.
- ❑ To describe how characteristics are inherited.
- ❑ To understand that more than one scientist was involved in discovering the structure of DNA.



Atoms and cells both contain a **nucleus**.

Why do you think these structures have the same name?



Keywords: nucleus, chromosomes, chromatid, centromere, genes, DNA, base sequences

Success Criteria		
Working towards	Working at	Working beyond
I can describe the shape and structure of DNA including the complementary base pairs.	I can describe how DNA is arranged into genes and stored as 23 pairs of chromosomes in the nucleus of a human cell.	I can describe genes as short sections of DNA that code for proteins, which determine specific inherited characteristics

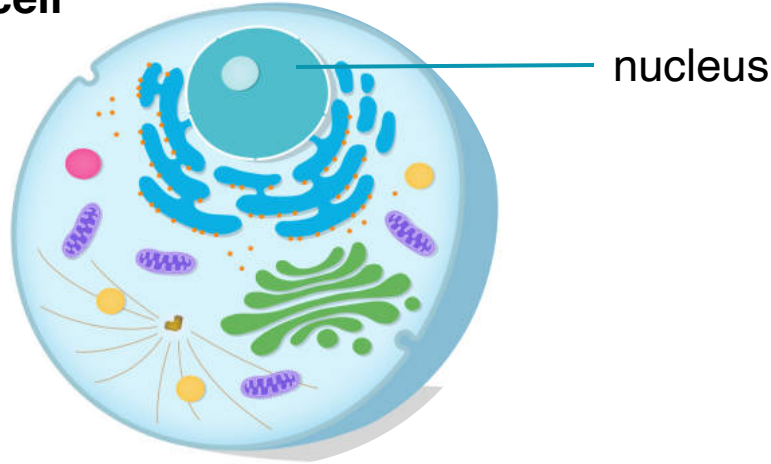
Chromosomes, genes and DNA

Success criteria

Lesson Title	Meeting	Exceeding	Mastering
Variation	☐ I can describe variation as the differences between individuals of the same species.	☐ I can explain the difference between genetic and environmental causes of variation and give examples of both	☐ I can distinguish between continuous and discontinuous variation and give examples of each
Chromosomes, genes & DNA	☐ I can describe the shape and structure of DNA including the complementary base pairs.	☐ I can describe how DNA is arranged into genes and stored as 23 pairs of chromosomes in the nucleus of a human cell.	☐ I can describe genes as short sections of DNA that code for proteins, which determine specific inherited characteristics
Gametes	☐ I can define gametes as sex cells involved in sexual reproduction and recognise sperm and egg cells as male and female gametes.	☐ I can explain why human gametes contain only 23 chromosomes and a zygote has 46.	☐ I can describe alleles as different versions of the same gene.
Inheritance	☐ I can explain that inherited characteristics are controlled by genes passed from parents to offspring via gametes.	☐ I can describe the difference between dominant and recessive alleles using examples.	☐ I can use genetic diagrams (Punnett squares) and pedigrees to predict inheritance patterns involving dominant and recessive alleles.
Natural Selection	☐ I can describe the process of natural selection as a mechanism for evolution.	☐ I can explain how variation, competition, and survival of the fittest lead to better-adapted individuals passing on their genes.	☐ I can use examples to illustrate how natural selection works in real populations.
Selective Breeding	☐ I can describe the process of selective breeding.	☐ I can explain how selective breeding can introduce new species over multiple generations.	☐ I am able to analyse the advantages and disadvantages of selective breeding and form my own opinion.

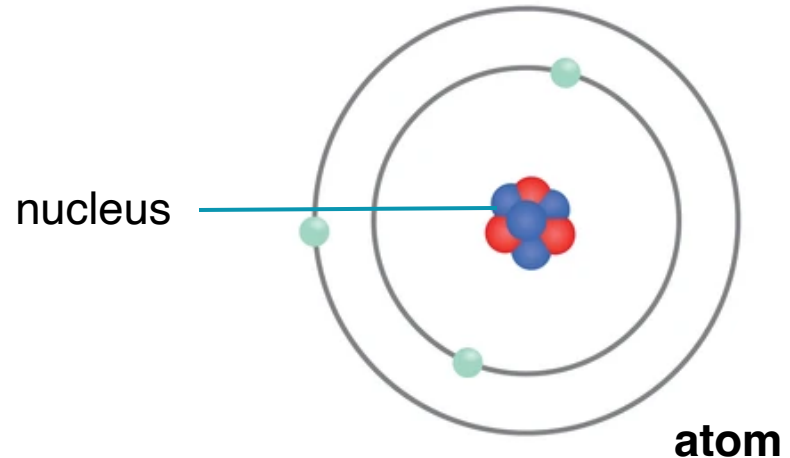
Nucleus

cell



The nucleus of a cell is much, **much larger** than the nucleus of an atom - it **controls the activities of the cell**.

Both are in the **centre** of a large structure.

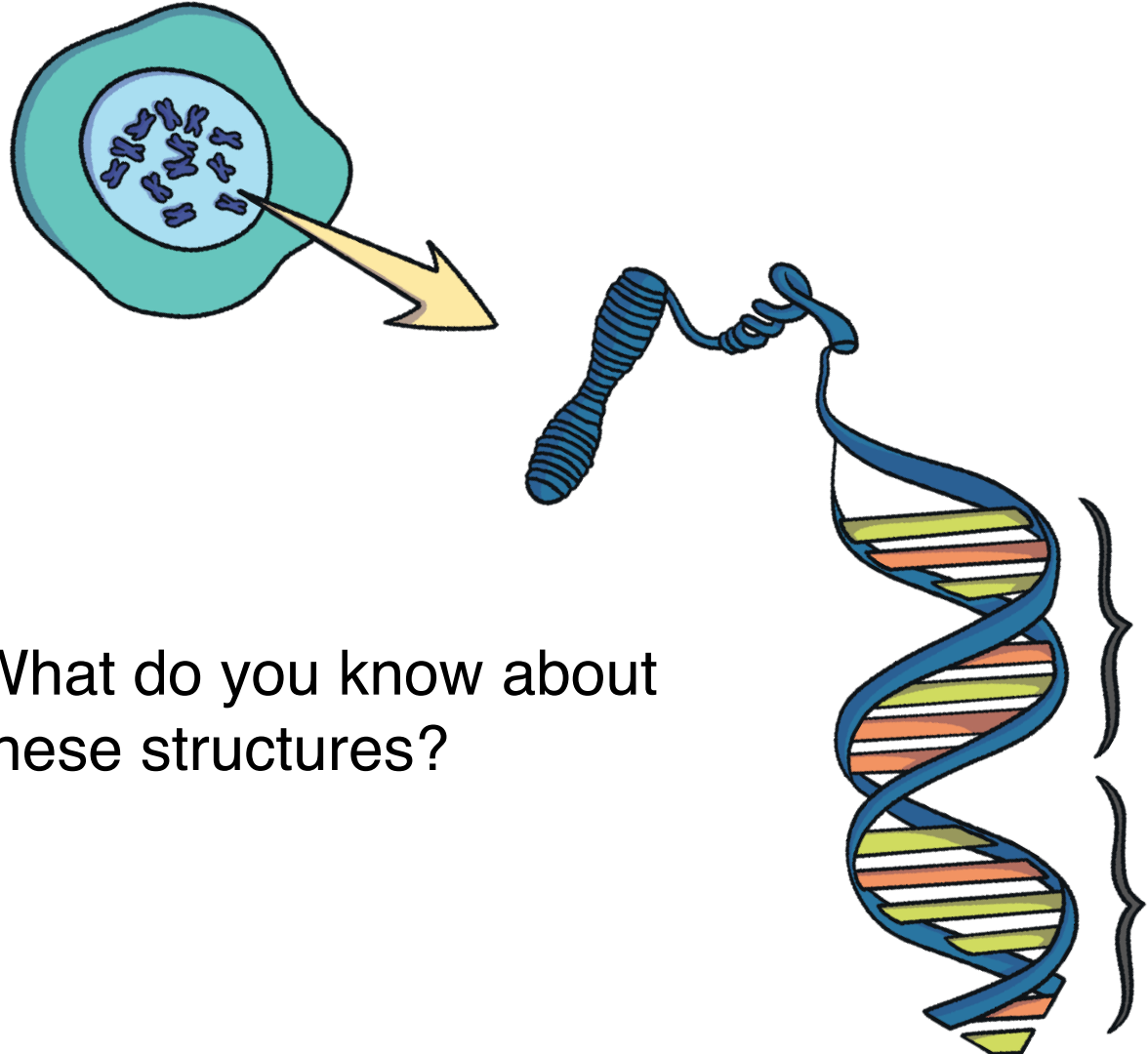


Nucleus

Chromosome

DNA

Gene



What do you know about these structures?

Human cells have 46 chromosomes (23 pairs)

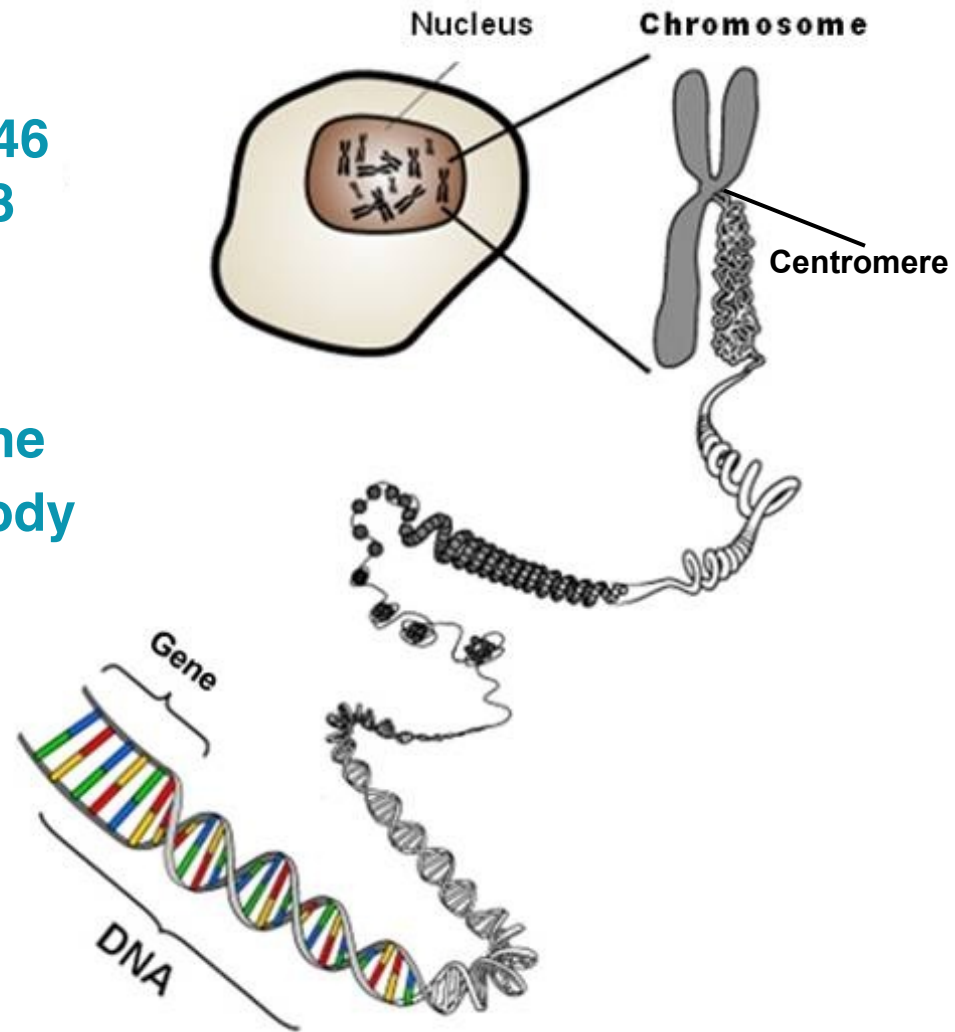
DNA contains all the instructions your body needs to live

Genes are the most basic unit of inheritance

Chromosome = long threads of DNA, found in the nucleus of a cell, which are made up of thousands of genes.

DNA = chemical substance that carries a code of instructions that determines how cells and organisms develop.

Gene = instructions, in the form of a DNA code, that affect many of the characteristics of an organism





1



2



3



4



5



6



7



8



9



10



11



12



13



14



15



16



17



18



19



20



21



22



or



X Y

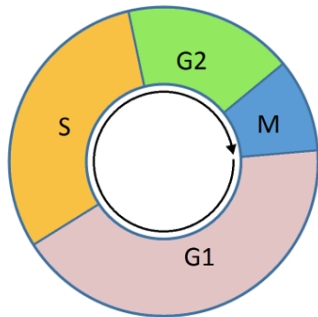
X X

Shown here are human chromosomes

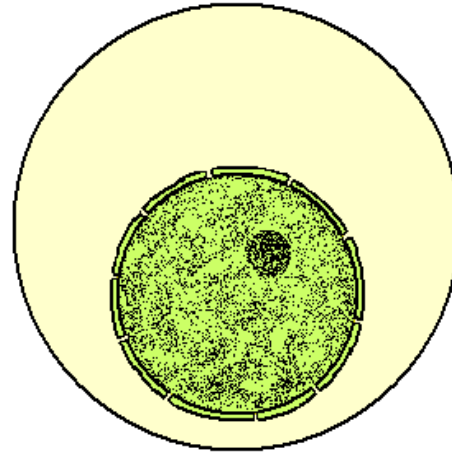
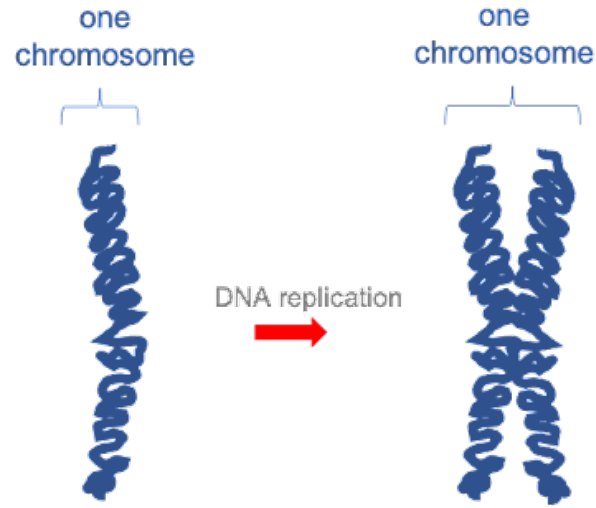
Chromosomes

We often see chromosomes having a 'cross' shape.

In reality, this is what a chromosome looks like when a cell has made a copy of itself, just before a cell divides.



G1 - Growth
S - DNA synthesis
G2 - Growth and preparation for mitosis
M - Mitosis (cell division)

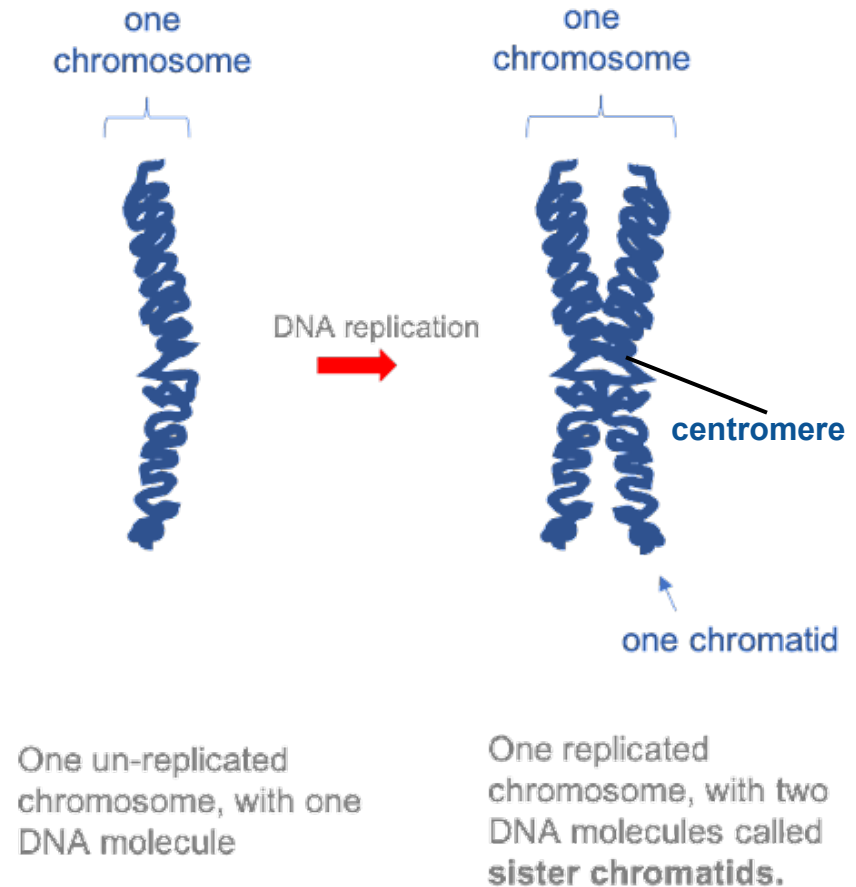


Chromosomes

We often see chromosomes having a 'cross' shape.

In reality, this is what a chromosome looks like when a cell has made a copy of itself, just before a cell divides.

The two copies (**chromatids**) are joined together by their **centromere**.



Chromosomes, genes and DNA... continued

Lesson objectives

- ❑ To explain what chromosomes are made up of. ✓
- ❑ To describe how characteristics are inherited.
- ❑ To understand that more than one scientist was involved in discovering the structure of DNA.



STARTER: What are chromosomes made up of?

Keywords: nucleus, chromosomes, chromatid, centromere, genes, DNA, base sequences

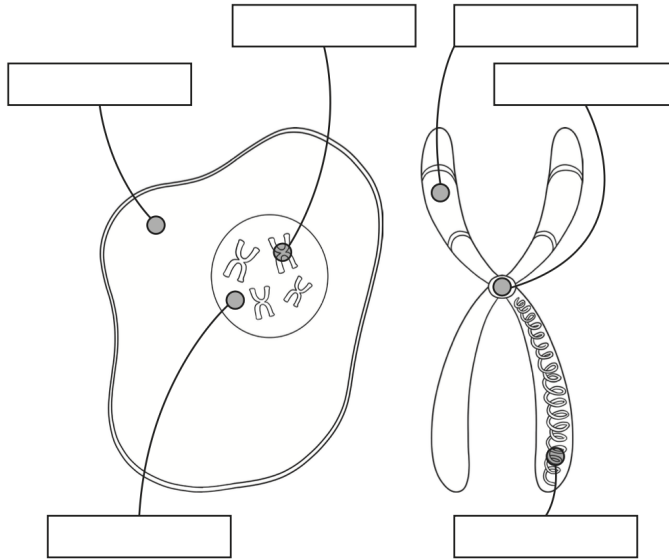
Chromosome structure



Chromosome Structure Labeling

Directions: Use the word bank below to label the parts of the chromosome.

chromatid	chromosomes	DNA
centromere	cell	nucleus

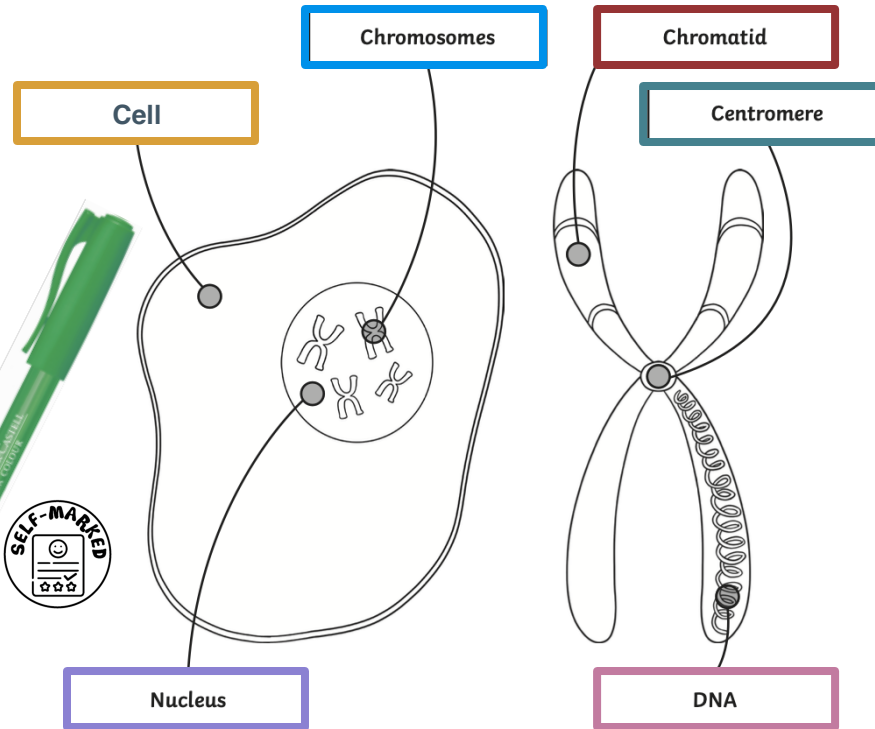


Next to/around the label, add the definition for each keyword.

Challenge: Write the definition, without looking at your notes!

Support: If English is not your first language, you are welcome to use your device to translate each label and write it next to it.

Chromosome structure



Cell: the smallest unit of life and the building blocks for all organisms

Nucleus: organelle that controls what happens in the cell

Chromosomes: long threads of DNA, found in the nucleus of a cell

Chromatid: one-half of a replicated chromosome

Centromere: structure that binds the two sister chromatids together

DNA: chemical substance that carries a code of instructions that determines how cells and organisms develop



DNA Double Helix 1953

"The secret of life"

For decades the Eagle was the local
pub for scientists from the nearby
Cavendish Laboratory.

It was here on February 28th 1953 that
Francis Crick and James Watson first
announced their discovery of how
DNA carries genetic information.

Unveiled by James Watson

25th April 2003

+ Franklin



Franklin & Wilkins

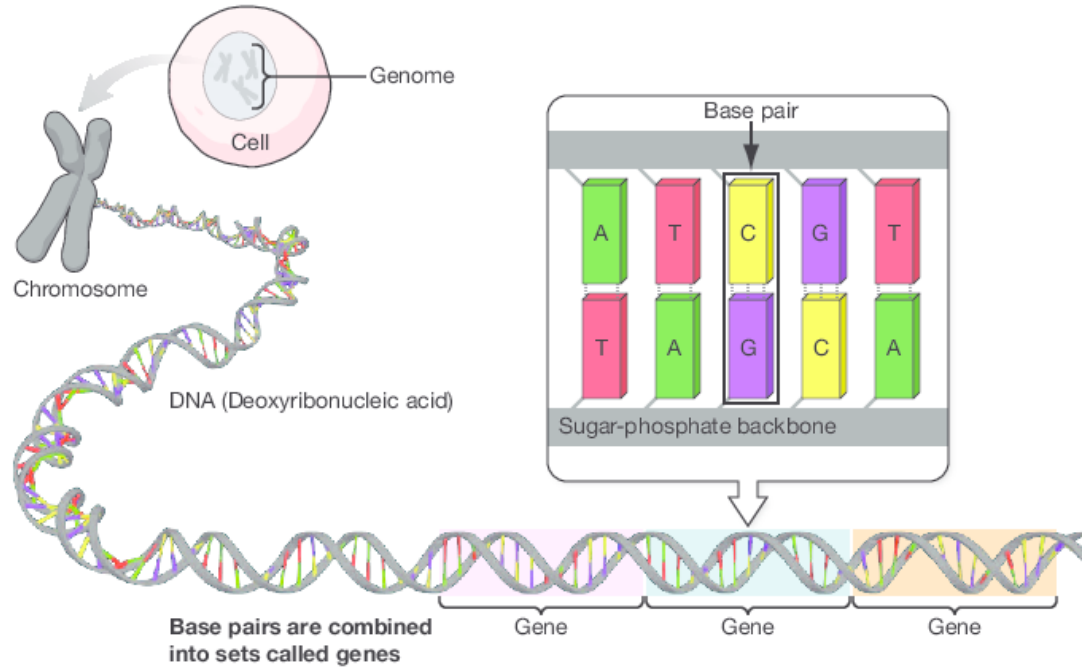
Watson & Crick

DNA and genes

Each chromosome is made up of a long tightly coiled DNA molecule.

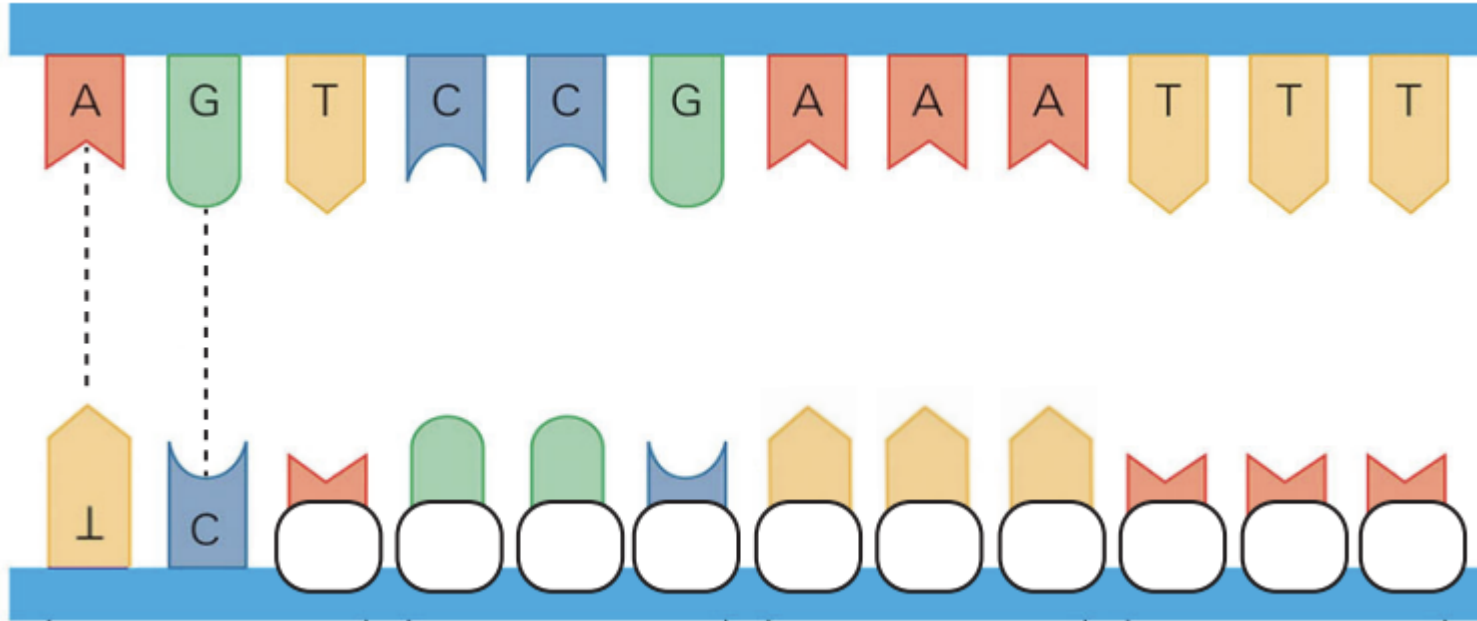
The DNA twists into a **double helix**.

The DNA is made up of a sequence of bases - **A**, **T**, **C** and **G**. Only **specific bases can pair up** and bond with each other. We call this **complementary base pairing**.



DNA and genes

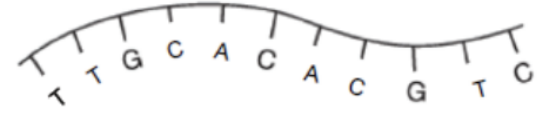
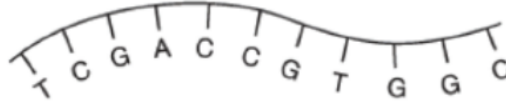
adenine (A) always bonds with thymine (T)
guanine (G) always bonds with cytosine (C)



Write down the missing base pairs in the boxes.

Most:

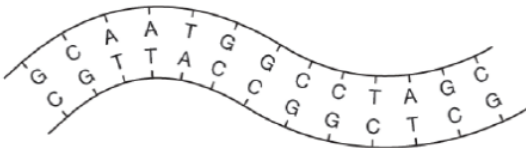
Draw the complementary DNA strand for the examples below.



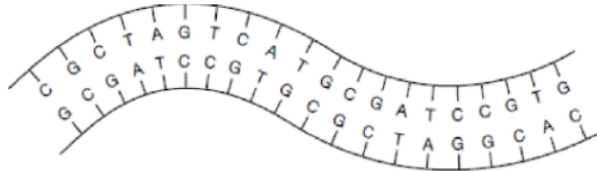
Some:

Identify the mistakes by circling them in the following sections of DNA.

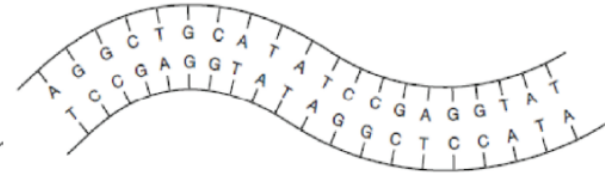
a

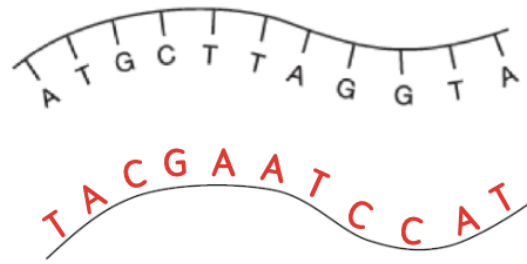
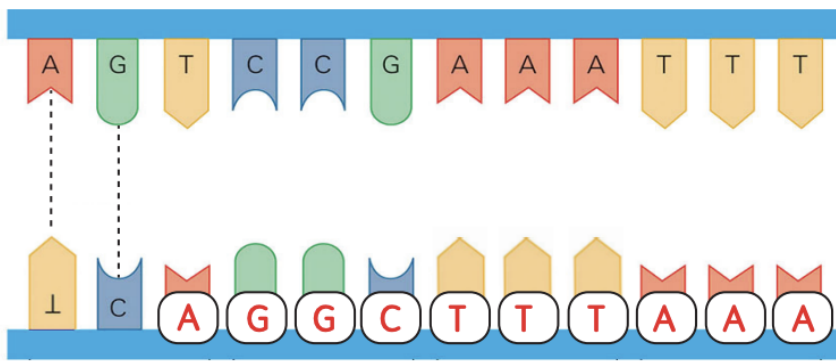


b

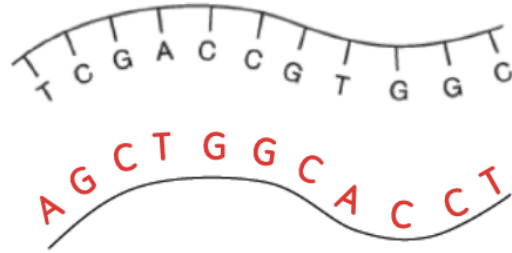


c

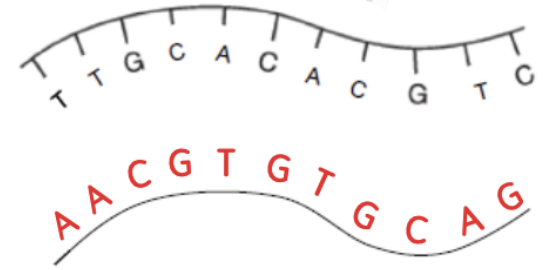




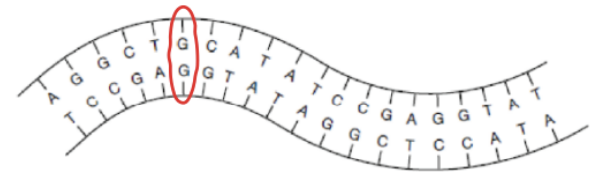
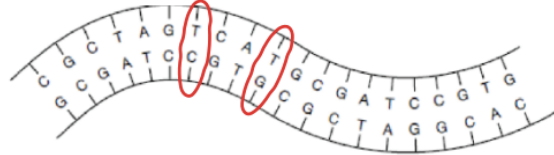
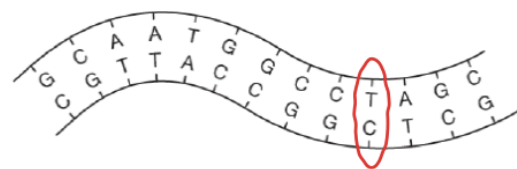
a



b



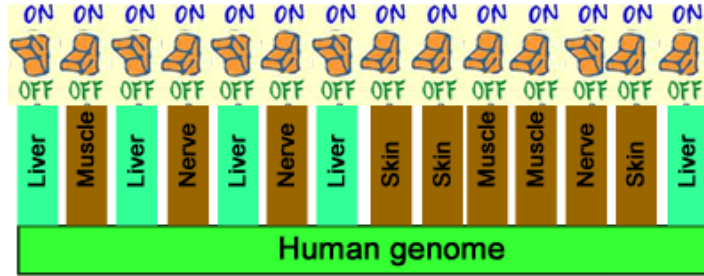
c



Every cell contains the same sequences of DNA!

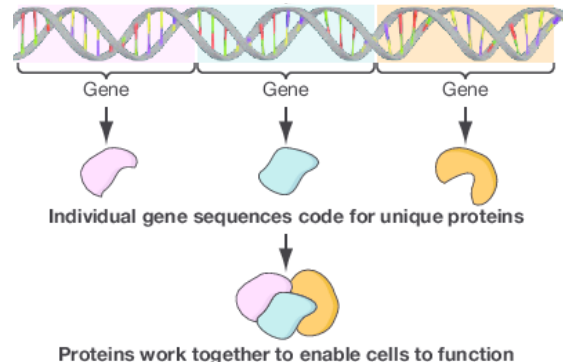
Cells have specific proteins that help them perform their functions.

Genes can turn proteins off or on.



Proteins allow cells in different organs to look and act differently from each other.

An organism inherits its genes from its parents. This makes them unique.

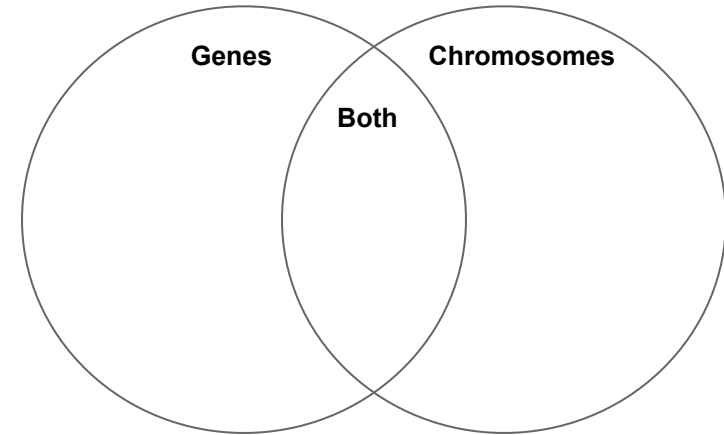


Chromosomes, genes and DNA

All: Use modeling clay to create models of chromosomes, DNA, and genes. Use whiteboard pens to label the desk.



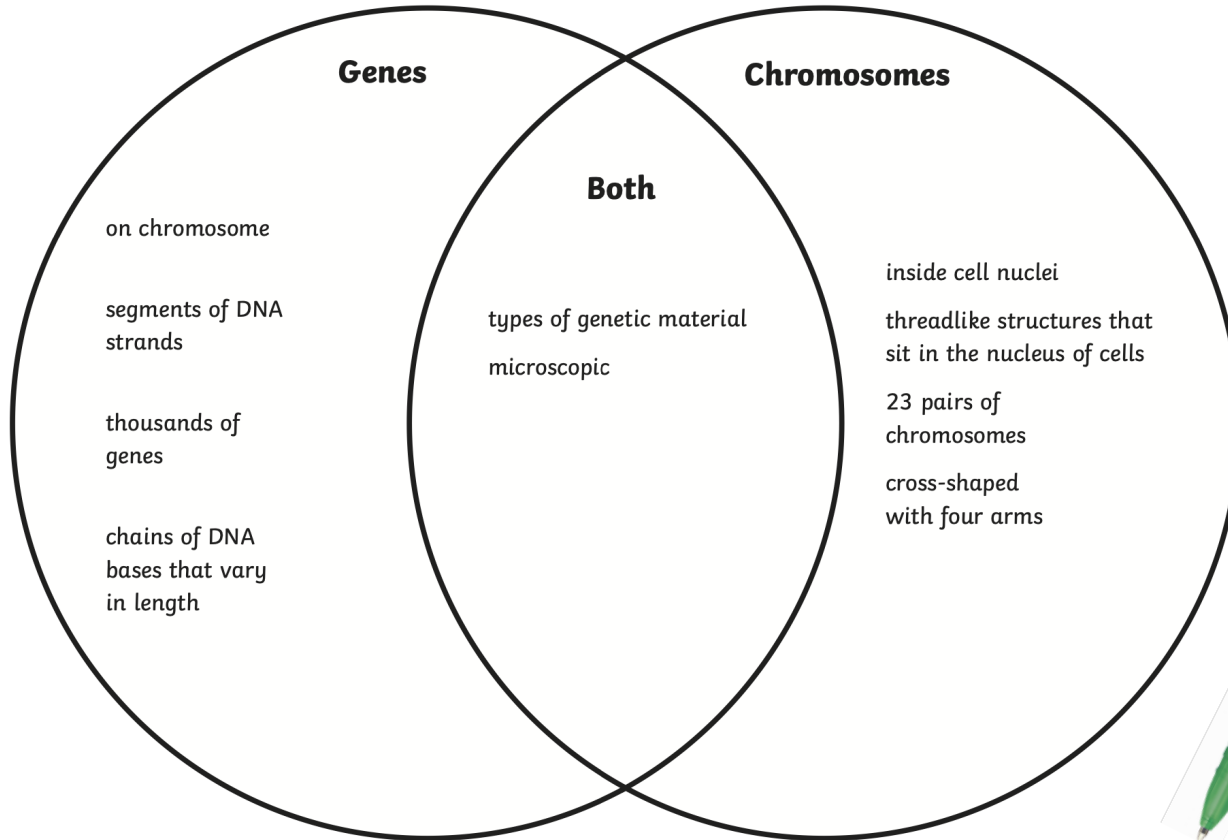
Most: Compare and contrast genes and chromosomes. Add brief notes and key words to complete the Venn diagram.



Some: Use the internet to research this question:
Do larger organisms have larger numbers of chromosomes in their cells?

Genes vs Chromosomes

Answers will vary but may include the following



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